

Mini Exercise 2 – XOR (Classification)

1. Using `tensorflow` and `keras`, train a neural network to predict the XOR truth table by Classification.
2. The inputs and outputs of the XOR table are shown below.

XOR gate truth table

Input		Output
A	B	A XOR B
0	0	0
0	1	1
1	0	1
1	1	0

3. Your network should accept two inputs (e.g. 0 and 1) and produce a single output value (e.g. 1).
4. The architecture of NN is:
 - (1) Two inputs (e.g., 0 and 1)
 - (2) Hidden layer with 4 neurons, activation function is tanh
 - (3) Output layer with 1 neuron, activation function is sigmoid
5. Use the `binary_crossentropy` as the loss function.
6. Train your network such that it consistently predicts a (correct) value that is 0 or a (correct) value that is 1, depending on the inputs.
7. Print the accuracy, and consider how to improve the accuracy.